

DPCN Assignment 1

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1 Team Name

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2 GitHub Link to Code

All code, figures and data are at https://github.com/shreyash-lohare/dpcn_a1_opinion_network. The repository is organised by pipeline stage — `src/preprocessing/` (sanitisation and imputation), `src/common/` and `src/analysis/` (noise floor, denoising, threshold selection), `src/communities/` (community detection and the reordered heatmap), and `src/viz/` (figures) — with ownership recorded per module in `CONTRIBUTIONS.md`. The whole analysis regenerates from a fresh clone with:

```
pip install -r requirements.txt
python run_all.py
```

`run_all.py` runs each stage in dependency order and names the responsible team member if an input artefact is missing.

3 Dataset Documentation

The questionnaire contains 60 items in four blocks of 15 — technology (T01–T15), education (E01–E15), society and ethics (S01–S15) and environment (V01–V15) — answered by 96 respondents. Five ordered categories are encoded symmetrically as $SD = -2$ through $SA = +2$, which represents the ordering without asserting equal spacing; a sixth option, No Comments (NC), is treated as missing. Internal consistency varies sharply by block: Cronbach’s α is 0.87 for environment and 0.82 for society but only 0.65 for technology and 0.58 for education.

3.1 The response ceiling

The most consequential property of this dataset is its ceiling. Figure 1 shows 79.8% of valid answers on the agree side, 43.1% Strongly Agree, and only 7.0% disagreeing. The grand mean is **+1.14** on the $[-2, +2]$ scale. Two consequences drive every later decision: item means barely discriminate, so the signal is in

which statements attract disagreement; and two respondents look similar under any raw similarity measure simply by both being agreeable.

The ceiling is not uniform. Environment ($\mu = 4.39$ on the 1–5 scale, 89.5% agree) and society ($\mu = 4.28$) are near consensus, while technology ($\mu = 3.99$) and education ($\mu = 3.91$, 13.8% disagree) retain real disagreement — and those two blocks also have the weakest internal consistency, consistent with their items tapping several distinct opinions rather than one settled disposition.

Ranked by disagreement, the twelve most contested items are seven education, four technology, one society, none environment. Compulsory attendance (E03, mean 2.06, 70.1% disagreeing) is the only statement the cohort actively rejects; written examinations (E02, 2.84) is next. The pattern is specific rather than oppositional — the same cohort agrees at 94.3% that curricula should be updated more often (E11). What is contested is assessment and teaching format, not the institution. Technology behaves likewise: teaching AI (T04) and AI as a learning aid (T02) score above 79%, while AI-assisted diagnosis (T08, 3.08) and autonomous vehicles (T09, 3.28) sit lowest. Since robots in hazardous work (T07) scores 82.4%, the objection is to delegating consequential judgment, not to automation.

3.2 Missing data and how it was recovered

Of 96 records, 85 are complete, 5 are entirely empty, and 6 stop partway. The five empty records are dropped, leaving 91 respondents; none is dropped merely for being incomplete. That leaves 205 structural blanks plus 37 NC cells, **242** in total. Dropout is monotone — a truncated tail, not scattered gaps — so later blocks rest on a slightly smaller base.

NC is not Neutral. Recoding a withheld answer as Neutral asserts a response nobody gave. A two-way additive model $\hat{x}_{iq} = \bar{\mu} + a_i + b_q$ fitted on the clean responses predicts 3.92 at the 37 NC cells against 3 for Neutral, so recoding would impose a systematic 0.92-point downward shift. NC use also shows no item- or respondent-level structure (item-uniformity $p = 0.27$; NC users’ mean 4.01 against 4.15, $p = 0.27$), so there is no basis for assigning it any fixed category.

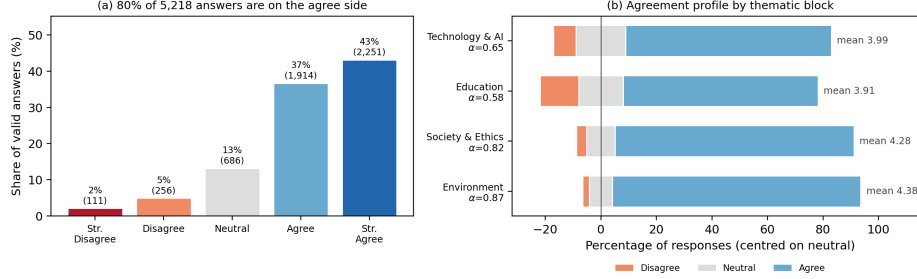


Figure 1: Response ceiling and block-level agreement. (a) All 5,218 valid answers: nearly four in five are on the agree side. (b) Agreement profile per block with block mean and Cronbach’s α . Environment and society are near consensus; education carries the most disagreement and the weakest consistency.

Table 1: Artificial-masking validation of the ordinal-regression imputer. The model beats a modal baseline under both mechanisms, by a wider margin where more co-answered items survive.

Masking	Ordinal model		Mode baseline	
	Acc.	MAE	Acc.	MAE
Scattered (NC-like)	.539	.534	.486	.626
Suffix (structural)	.498	.612	.482	.655

Recovery by ordinal regression. For target item j the other 59 items form the predictor vector, and we fit one regularised proportional-odds ordinal logistic regression per item, with latent score $z_i = \beta_0 + \beta^\top \mathbf{x}_{i,-j}$ and ordered thresholds $\theta_1 < \dots < \theta_4$; the prediction is the maximum-probability category. This respects the ordering of the scale without assuming equal intervals. Predictors that are themselves missing are filled with the training-fold column median, computed per fold to prevent leakage, and a ridge penalty controls the 59-dimensional fit.

Does it work? The true values are unavailable, so we validate by artificial masking under both mechanisms: scattered masking hides 37 observed cells inside complete respondents (40 trials), suffix masking hides a contiguous tail after a simulated stopping point (15 respondents per trial, 15 trials). Because the target is ordinal, MAE matters more than exact accuracy: being one category out is not the same error as being four. Table 1 reports both.

One caution carries forward. Comparing four NC treatments at $\tau = 0.30$, the correlation matrices agree at $r > 0.94$ while their community assignments agree at AMI as low as 0.337 (dropping NC users, which discards 17 respondents to remove 37 cells). Stability of a correlation matrix is not evidence that a partition of it is stable — a warning that Section 6.2 makes quantitative.

3.3 Row-centring

Imputation yields a complete 91×60 matrix, but not one that can be correlated directly: with a grand mean of +1.14, a respondent who agrees with everything raises all 60 entries together, which reads as association between every pair of items at once. We replace each entry by its deviation from that respondent’s own mean,

$$x'_{i,j} = x_{i,j} - \bar{x}_{i.}, \quad \bar{x}_{i.} = \frac{1}{60} \sum_k x_{i,k}. \quad (1)$$

Centring is applied strictly after imputation and strictly before correlation. Note that rank correlation does not remove this bias automatically: Spearman is invariant to a monotone transform of a column, not to an additive shift that differs by row. Graph 3 (Section 5.3) measures how much it matters.

4 Pipeline Followed

4.1 What the network is

Vertices are the 60 questionnaire items, and edges are signed correlations between items, computed on the row-centred matrix and retained with their sign. Edges are weighted by default and binarised only where a metric requires it (path lengths, triad counts).

Items were chosen over respondents deliberately: an item network is thresholded directly on a measured quantity and every vertex carries an interpretable label, whereas a respondent network has no directly measured edge and would have to be constructed by an imposed neighbourhood rule. Signs are kept because a negative edge — two questions dividing the cohort in opposite directions — is a stronger statement than mere association; Graph 8 (Section 5.8) shows this sign pattern separating topic blocks cleanly once acquiescence is removed.

4.2 Step by step

1. Parse and encode. Identify the 60 item columns, map the five categories to $\{-2, \dots, +2\}$ and NC to NaN. Yields 96×60 with 542 missing cells.
2. Filter respondents. Drop only the five entirely empty records. 91×60 , 242 missing cells.
3. Impute (Section 3.2). One regularised proportional-odds ordinal logistic regression per item, predicting it from the other 59.
4. Row-centre (Section 3.3). Subtract each respondent's own mean across all 60 items — strictly after imputation, strictly before correlation.
5. Correlate. Pearson correlation over the 60 columns, giving a 60×60 signed matrix with $\binom{60}{2} = 1,770$ distinct off-diagonal entries.
6. Establish the noise floor (Section 5.4). Re-run steps 4–5 on 200 independently column-permuted copies of the matrix from step 3.
7. Denoise (Section 5.5). Keep only the eigenvalue modes exceeding the Marchenko–Pastur upper edge λ_+ , rebuild the matrix from those modes, and restore the unit diagonal.
8. Threshold (Section 5.7). Sweep all 46 cutoffs from 0.05 to 0.50 and select by a stated rule rather than by inspecting the result. Selects $|r| > 0.22$.
9. Measure (Section 5). Community detection (Graphs 6 and 9) and topic-block connectivity (Graph 8) — each compared against an explicit null.

5 Analysis and Visualisations

This section reports the nine analyses. Each states what the figure shows, what was found, and what follows from it.

A note on respondent counts. All figures below report $n = 91$: every non-empty respondent is retained and the missing cells are recovered by ordinal regression (Section 3.2). An alternative pipeline that instead drops respondents with more than 12 missing items, leaving 86, is examined in Section 6.2.

5.1 Graph 1: Missingness map and response distribution

Figure 2 shows every cell of the raw 96×60 matrix as missing or present, respondents sorted by missing count, beside the distribution of per-respondent means.

Findings. The 542 missing cells (505 structural blanks plus 37 NC) are extremely unevenly distributed: five entirely empty respondents account for 300 of them, and the remaining 242 spread thinly across the other

91 — a rate of 4.4%. Below the cutoff the map is almost clean, with two visible tails from respondents who stopped early. The right panel locates the grand mean at 1.14, with only one respondent below zero.

Implication. This justifies the retention policy directly: all but five respondents answered at least 80% of items, so there is no population of badly incomplete respondents whose removal would simplify anything, and a count-based exclusion rule would discard usable respondents to remove very few cells. The right panel is the first direct evidence of the acquiescence problem that row-centring corrects.

5.2 Graph 2: Item variance ranking

Figure 3 ranks all 60 items by standard deviation on the uncentred matrix. Centring is deliberately not applied here: it changes each item's variance and would answer a different question (spread after removing each respondent's baseline) than the one asked (how divided the cohort is on the statement as posed).

Findings. Item SD ranges from 0.50 (E15, near-unanimous) to 1.18 (E04, online learning). The five most contested are E04, E03, E02, T08, T07. By block mean SD: technology 0.92, education 0.88, society 0.79, environment 0.72.

Implication. Technology has the highest average variance, but the sharpest disagreements are educational — three of the five most divided items are E02–E04, all about assessment and teaching format — and environment contributes none of the top ten. This is a property of the instrument, not of preprocessing: the questionnaire contains one settled domain, one moderately settled, and two live ones. Consensus items carry little information for a correlation network, since an item everyone endorses cannot covary with anything, so the signal is necessarily concentrated in the contested tail.

5.3 Graph 3: The effect of row-centring

Figure 4 overlays all $\binom{60}{2} = 1,770$ pairwise item correlations before and after row-centring, on identical bin edges.

Findings. The raw distribution is displaced positive (mean and median both 0.151), with almost no pair appreciably negative. After centring it sits on zero (mean -0.004 , median -0.008) with positive skew 0.221, leaving a heavier right tail of genuinely co-varying pairs. At $|r| \geq 0.30$ the surviving pairs fall from **327** to **42** — a reduction of 87%.

Implication. This converts the centring argument from an assertion into a measurement: seven eighths of the apparent association in the raw matrix is shared response style. Built on raw scores the network would have been dense, almost entirely positive, and would

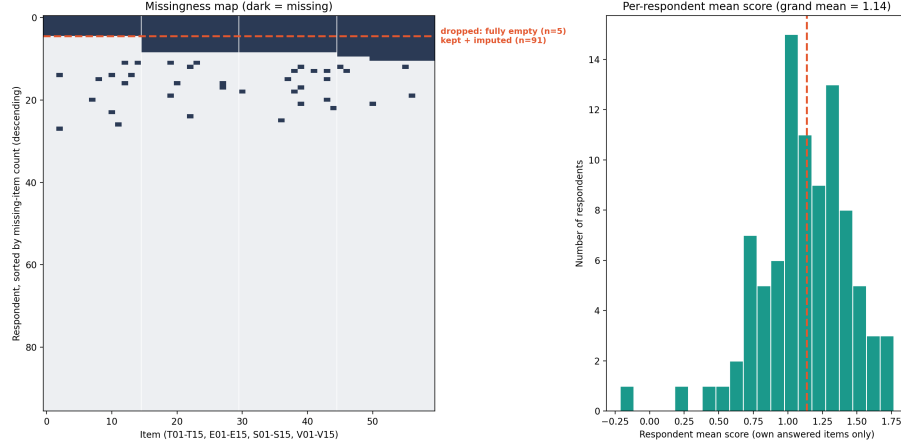


Figure 2: Missingness map and per-responder response means. Left: missingness over all 96 raw respondents, sorted by missing-item count. The dashed line marks the exclusion rule — only the five entirely empty records are dropped, and the remaining 91 have their 242 missing cells imputed. Right: per-responder mean over answered items, grand mean 1.14.

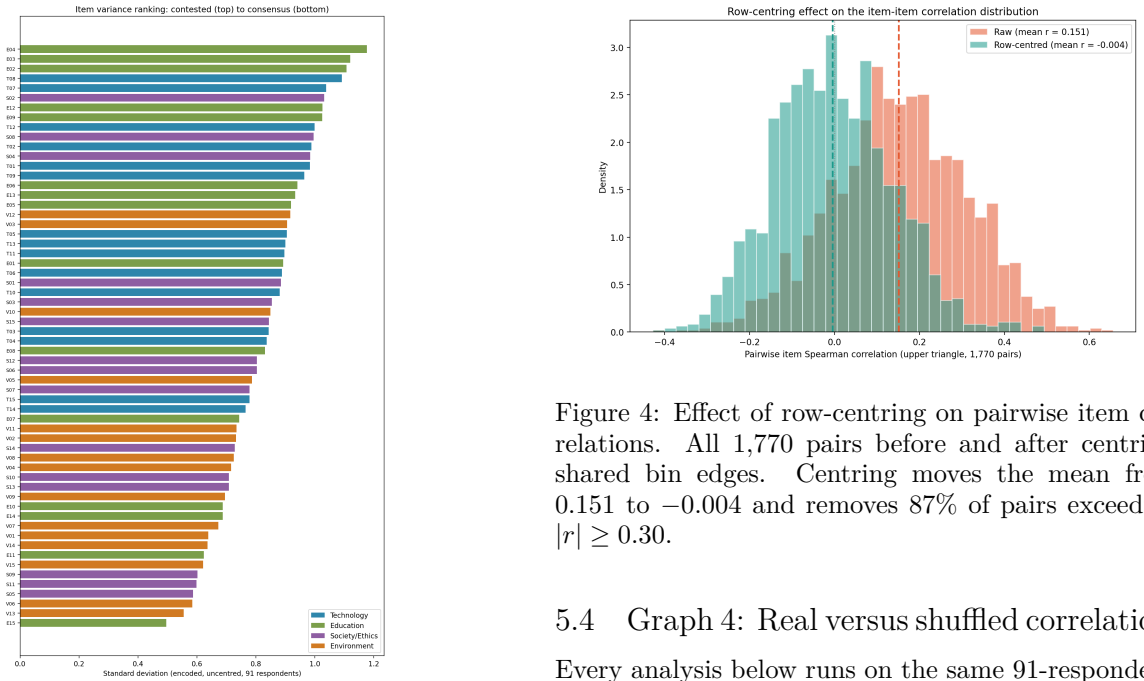


Figure 3: Item variance across the four survey blocks. All 60 items by standard deviation on the uncentred matrix, coloured by block. Contested items concentrate in education and technology; environment clusters at the consensus end.

have supported confident but spurious conclusions. The emergence of negative correlations after centring — invisible in the raw distribution — is also what makes the sign pattern in Graph 8’s topic-block analysis interpretable at all.

Figure 4: Effect of row-centring on pairwise item correlations. All 1,770 pairs before and after centring, shared bin edges. Centring moves the mean from 0.151 to -0.004 and removes 87% of pairs exceeding $|r| \geq 0.30$.

5.4 Graph 4: Real versus shuffled correlations

Every analysis below runs on the same 91-responder, ordinal-imputed matrix, so no part of this report silently uses a different sample.

Figure 5 compares the 1,770 observed item–item correlations against a null that destroys inter-item association while preserving every item’s marginal exactly. Each of 200 replicates permutes every column independently and re-runs the full pipeline — row-centre, then correlate — on the shuffled matrix. This detail is not cosmetic: one permutation applied to all columns, or permuting rows, merely relabels respondents and leaves the correlation structure numerically untouched.

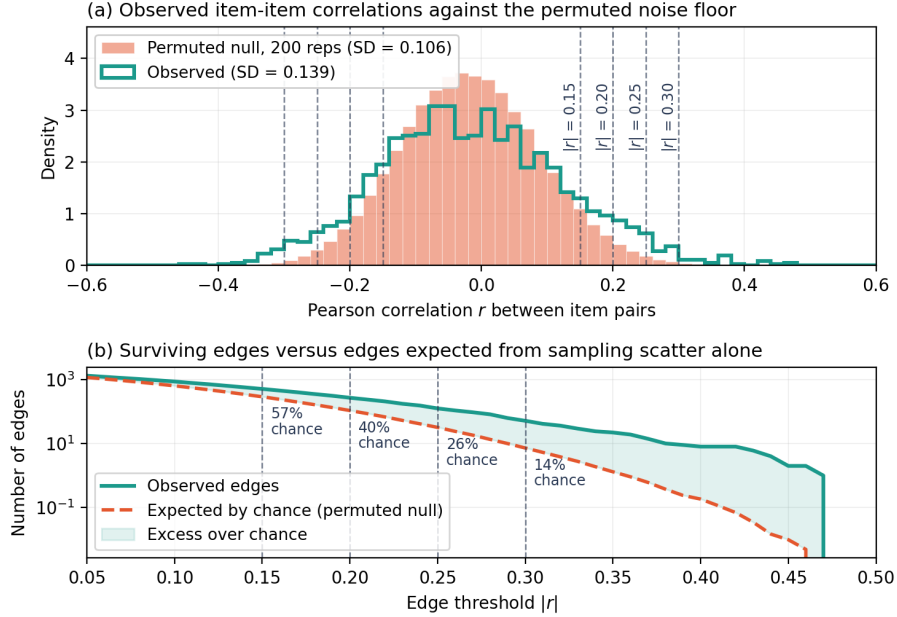


Figure 5: Observed versus permuted item correlations. (a) Observed correlations against a 200-replicate permutation null that re-runs the whole pipeline on independently shuffled columns; SD 0.139 against 0.106, the excess confined to the tails. (b) Surviving edges against edges expected from sampling scatter alone (log scale); the shaded band is the excess over chance.

Findings. Correlations from $n = 91$ respondents scatter about zero with SD $1/\sqrt{n-3} = 0.107$ even when nothing is related. The permuted null reproduces this at **0.106**, against **0.139** observed. Real structure exists but is modest, and lives in the tails:

$ r >$	0.15	0.20	0.25	0.30
Observed edges	502	266	124	51
Expected by chance	286	107	32	7
Chance fraction	57%	40%	26%	14%

Implication. At the conventional cutoff $|r| > 0.15$ more than half the surviving edges are what 1,770 simultaneous estimates from 91 people produce by luck. A network drawn there is mostly noise, and no downstream community detection can separate the two after the fact. This makes threshold selection a measured trade-off rather than a convention, and supplies the chance curve that Graph 7 is read against.

5.5 Graph 5: Eigenvalue spectrum against Marchenko–Pastur

Graph 4 establishes that a noise floor exists but not how many genuine dimensions sit above it. PCA cannot settle this: PC1 explains only 8.3% of variance, equally consistent with weak structure and with none. Marchenko–Pastur theory answers it exactly — for

a correlation matrix from T observations of N pure-noise variables the eigenvalues fall inside an analytically known band, and anything above its upper edge carries more structure than T observations could produce by chance.

Findings. With $T = 91$, $N = 60$, $Q = 1.517$ and $\lambda_+ = (1 + 1/\sqrt{Q})^2 = \mathbf{3.283}$. The bulk of the spectrum tracks the theoretical density closely and exactly four eigenvalues clear the band — 4.38, 4.06, 3.55, 3.35 — carrying **25.6%** of variance (Figure 6). The count is comfortable rather than marginal: the fourth clears by 2.0% while the fifth, at 2.86, sits well inside the band.

Reading the loadings, the four modes are (i) environmental and social commitment against technology governance and assessment tradition, (ii) developmental optimism against regulatory caution, (iii) AI-enabled education and industry against environmental stewardship, and (iv) systemic technological action against individual and community obligation. Modes (i) and (iii) split cleanly along topic blocks; (ii) and (iv) cut across them and are offered as readings of the loadings rather than established constructs.

Implication. Three quarters of this correlation matrix is sampling noise. That disciplines everything downstream: the network is rebuilt from the four significant modes alone, with the diagonal restored to unity, so later analyses operate on the part of the matrix that 91 respondents can actually support.

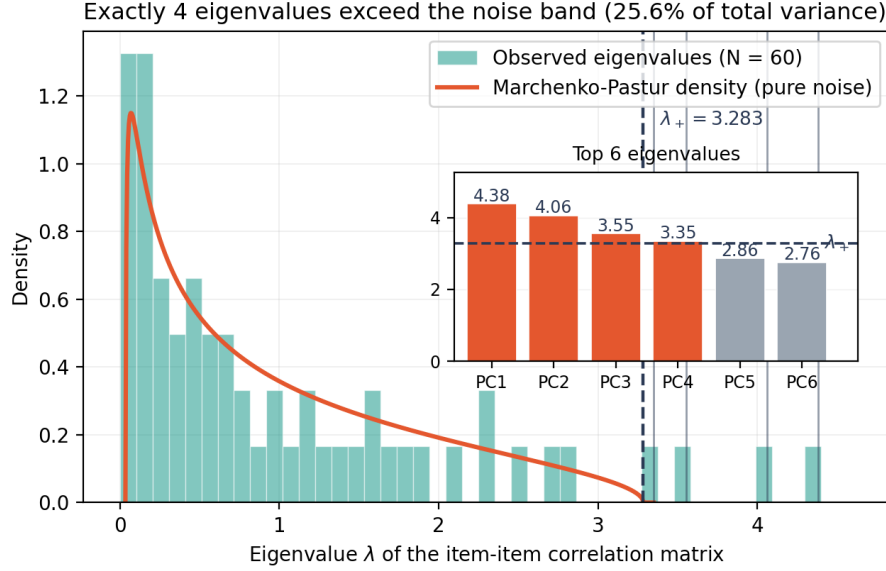


Figure 6: Eigenvalue spectrum against the Marchenko–Pastur noise band. The spectrum of the row-centred item correlation matrix against the Marchenko–Pastur density for pure noise of the same shape ($T = 91$, $N = 60$). Exactly four eigenvalues exceed $\lambda_+ = 3.283$, carrying 25.6% of variance. Inset: a clear gap between the fourth (3.35) and fifth (2.86).

5.6 Graph 6: Reordered correlation heatmap

Figure 7 presents the MP-denoised correlation matrix reordered by the community assignments detected in Graph 9 (Section 5.9). A diverging colour map centred at zero distinguishes positive from negative correlations, and the marginal colour strips mark the predefined thematic blocks (technology, education, society/ethics, environment).

Findings. The block-diagonal structure of the reordered matrix confirms that the community detection groups densely connected items. The marginal strips, however, show those data-driven communities cutting directly across the survey’s predefined thematic blocks, and the formal comparison yields a normalised mutual information (NMI) of 0.205.

Implication. Respondents’ association patterns do not partition neatly into the four hypothesised domains. The partition is a good description of the graph, but it is not the partition the instrument was built around.

5.7 Graph 7: Threshold and percolation sweep

Reporting a network at one threshold invites the objection that the threshold was chosen to produce the result. Figure 8 answers it by rebuilding the signed graph at all 46 cutoffs from 0.05 to 0.50 and tracking edge count, giant component, fragmentation and Louvain modularity across the range, so the chosen value is visibly a point on a curve. The sweep also absorbs

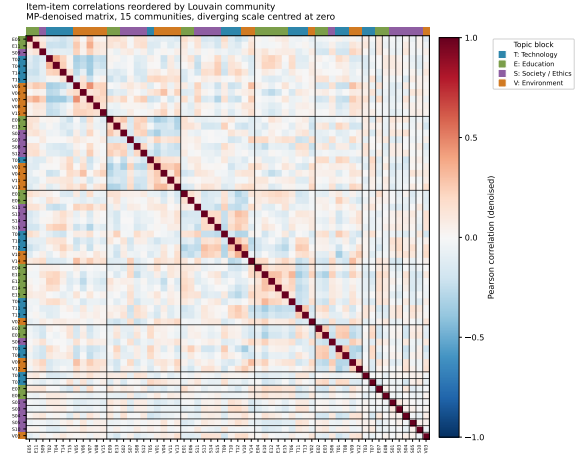


Figure 7: MP-denoised item correlation matrix reordered by community. The 15 communities are separated by black lines, with topic blocks marked in the margins. The communities cut across the predefined topic blocks, showing that respondents’ association patterns do not follow the survey’s own design.

the giant-component analysis entirely.

Findings. The two constraints pull opposite ways: raising the threshold lowers the chance fraction from Graph 4, but the graph eventually shatters — by $|r| > 0.27$ the giant component holds under half the items. We therefore state the rule explicitly: the highest cut-

Threshold sweep on the MP-denoised item network

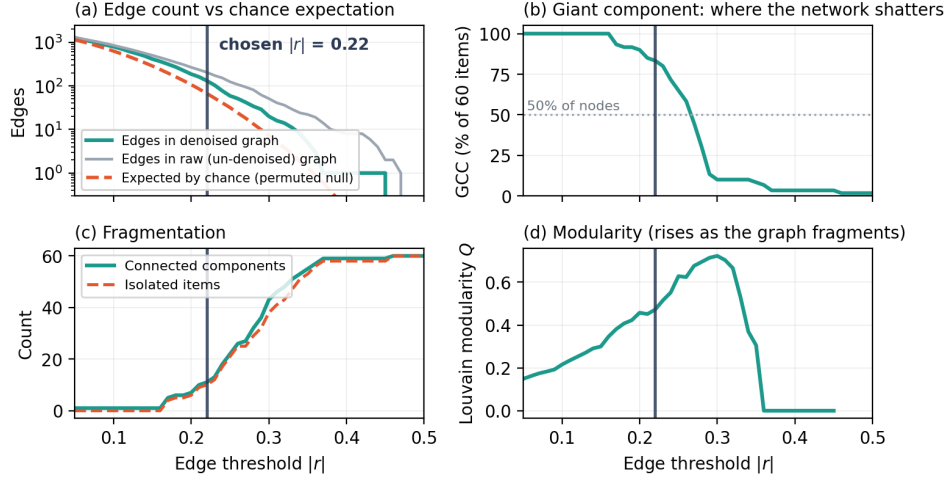


Figure 8: Threshold sweep and network fragmentation. The signed item network rebuilt at all 46 cutoffs from 0.05 to 0.50. (a) Edge count against the chance expectation from Graph 4. (b) Giant component. (c) Components and isolates. (d) Louvain modularity, which reaches 0.72 only once the graph has broken into 43 pieces. The vertical line marks the selected $|r| > 0.22$.

off at which the giant component still spans more than two thirds of the items and mean degree remains above 4, so community and path metrics stay meaningful. Since spectral denoising has already removed the sampling noise the chance column measures, raising the cutoff further would penalise the same noise twice. This selects $|r| > 0.22$: 131 edges (68 positive, 63 negative), mean degree 4.37, a giant component of 50 of 60 items (83%), modularity $Q = 0.472$. At the next step, $|r| > 0.23$, mean degree falls to 3.47 and the rule stops being satisfied.

Two independent controls agree there despite sharing no machinery. The permutation null implies the raw graph’s 205 edges contain about 67 chance edges, leaving roughly 138 real; spectral denoising, which never sees the null, retains 131 — a 5% difference.

Implication. Panel (d) is the cautionary one. Modularity climbs steadily to 0.72 as the threshold rises, but by then the graph is 43 components with 38 isolated items. Quoting a single high modularity without the sweep beside it would report fragmentation as community structure — the exact error the null models of Section 6.2 are designed to catch.

5.8 Graph 8: Topic-block connectivity

The questionnaire arrives with its own partition — fifteen items each for technology, education, society and environment — which is an editorial claim about what belongs together. Whether it is also a statistical one

should be asked before any community result is interpreted against it. We collapse the 60×60 correlation matrix into a 4×4 block matrix, averaging the 105 distinct within-block pairs on the diagonal (self-correlations excluded: keeping the unit diagonal would manufacture the very block structure being tested). Significance comes from a block-label permutation — the matrix is held fixed and the T/E/S/V labels reshuffled over 5,000 seeded draws — which needs no threshold and is immune to any constant offset in the matrix.

Findings. Row-centring is what makes the design visible at all. On the raw matrix (Figure 9, left) all sixteen cells are positive (+0.041 to +0.377): acquiescence makes every topic look connected to every other. After centring, the sign pattern separates cleanly — every diagonal cell positive, every off-diagonal negative. Within-block mean $r = +0.0746$ against between-block -0.0282 gives a gap of **+0.1027** against a null of -0.00003 ± 0.0077 : $z = 13.4$, with no permutation in 5,000 reaching the observed value ($p < 0.0002$).

The blocks are not equally coherent: environment +0.136, technology +0.066, education +0.054, society/ethics +0.042. Environment is more than three times as internally consistent as ethics. S–V is the one seam, at -0.004 the only pair not clearly separated and the strongest off-diagonal cell in the raw matrix (+0.256).

Implication. The thematic design is statistically real, so a later community that aligns with a block is recovering something the instrument built in, while one that cuts across blocks is the more informative

Graph 9: topic-block connectivity, 91 respondents x 60 items

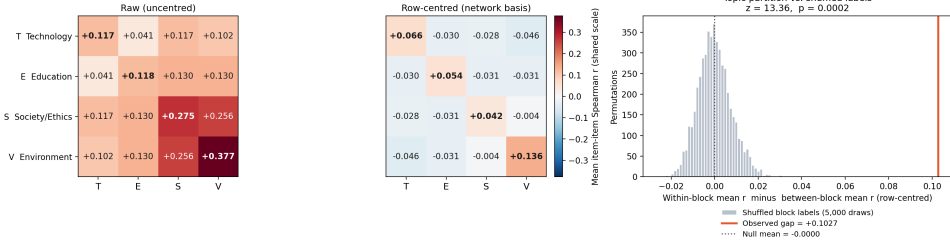


Figure 9: Topic-block connectivity before and after row-centring. Left: mean item-item correlation within and between blocks on the raw matrix — every cell positive, the design invisible under acquiescence. Centre: the same collapse after row-centring, shared colour scale; diagonal positive, all off-diagonals negative. Right: null distribution of the within-minus-between gap under 5,000 reshufflings of the block labels. The observed gap +0.1027 ($z = 13.4$) is not approached by any permutation.

object — which is exactly what Graph 9 finds (NMI 0.205). The uneven diagonal sharpens this: environmental attitudes travel as a package, whereas ethics items are close to independent, so respondents hold ethical positions issue by issue rather than as a doctrine. One caveat: the negative off-diagonals do not mean topics oppose one another. Row-centring forces each row to sum to zero, pushing the grand mean of all 1,770 correlations to -0.0038 ; the values should be read relative to the diagonal, not as opposition.

-3.14 , $p = 1.000$. Scrambled data pushed through the identical pipeline produces more modular graphs than the real data does. Section 6.2 takes this apart: the modularity is manufactured by thresholding a correlation matrix estimated from 91 respondents, not carried by the opinions themselves.

5.9 Graph 9: Communities against null models

Figure 10 compares three community-detection algorithms on the item network at $|r| > 0.22$, and Figure 11 then tests the resulting partition against three nulls: degree-preserving rewiring, an Erdős-Rényi graph, and a column-permutation null that scrambles responses independently per item.

Which algorithm. We began with Louvain, which found 15 communities at $Q = 0.472$. Louvain has a documented flaw: its greedy aggregation phase can leave communities internally disconnected, capping partition quality. Leiden adds a refinement phase that guarantees well-connected communities and found a strictly better partition, $Q = 0.491$, again at $k = 15$. Girvan-Newman, a divisive method that repeatedly removes the highest-betweenness edge, peaks at $Q = 0.447$ — also at $k = 15$ — and then declines monotonically. That all three strategies converge on the same 15 communities makes the partition size a robust feature rather than an artefact of one optimiser, so Leiden is used for the primary analysis.

And then the nulls disagree. Leiden's $Q = 0.491$ clears the Erdős-Rényi null (0.420 ± 0.016 , $z = 4.46$) and the rewiring null (0.333 ± 0.013 , $z = 12.13$) decisively. Against the column-permutation null it does not merely fail, it falls below it: 0.629 ± 0.044 , $z =$

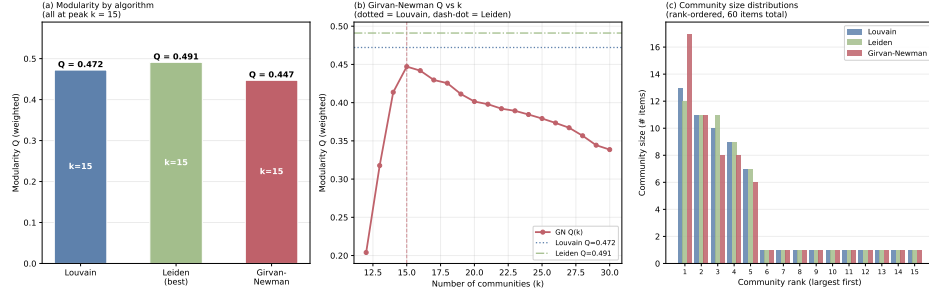


Figure 10: Community detection algorithms compared on the item network. (a) Peak modularity per algorithm; all converge on $k = 15$, Leiden highest at $Q = 0.491$. (b) The Girvan–Newman $Q(k)$ curve rising to a clear peak at $k = 15$ before declining, with Louvain and Leiden peaks marked. (c) Rank-ordered community sizes, similar across all three methods.

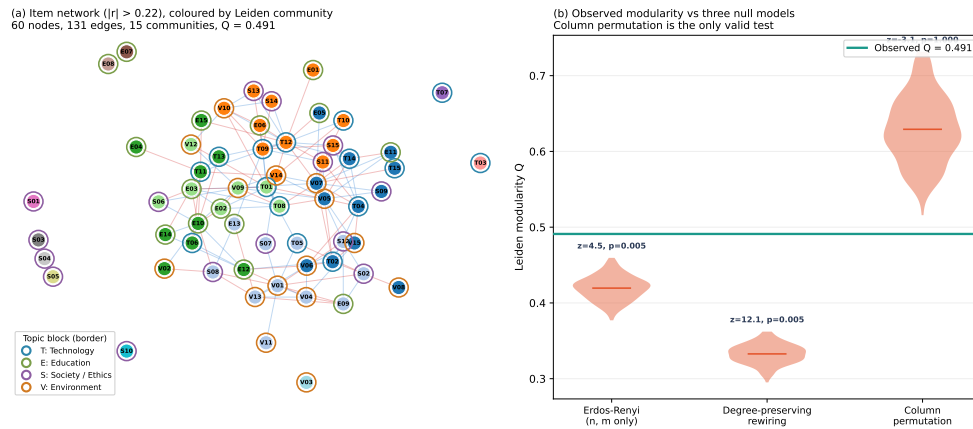


Figure 11: Item-network community structure under three null models. (a) The item network at $|r| > 0.22$, nodes coloured by Leiden community, borders marking the predefined thematic block. (b) Observed modularity $Q = 0.491$ sits comfortably above the Erdős–Rényi and degree-preserving rewiring nulls but below the column-permutation null that reproduces the construction.

6 Results and Discussion

6.1 What the data supports

Superficial consensus, genuine local structure. The headline is not polarisation. Agreement runs at 79.8% across every block and PC1 explains about 8% of variance, so there are no two opposed camps. There is a broad agreeable consensus with real disagreement concentrated in identifiable places.

Disagreement is domain-specific, not dispositional. Environment and society are settled; technology and education are not. Within education the cohort is precise about what it rejects — compulsory attendance and written examinations — while endorsing curriculum reform above 94%. Within technology the objection tracks delegated consequential judgment (diagnosis, autonomous vehicles) rather than automation as such: robots in hazardous work still score 82.4%.

Most apparent structure was response style. Row-centring removes 87% of the item pairs exceeding $|r| \geq 0.30$. Any analysis skipping this step finds dense, uniformly positive structure and is measuring agreeableness.

Communities ignore the survey’s own design. The thematic blocks are statistically real ($z = 13.4$ on the within-minus-between gap, Graph 8), yet the 15 detected communities align with them only at $\text{NMI} = 0.205$. Respondents associate ideas across topic boundaries, not within the categories the questionnaire was built from.

6.2 What failed, and what that tells us

Three results did not come out the way the analysis was set up to expect. They are reported because each one changes what may legitimately be claimed.

1. Most of the correlation matrix is noise. Marchenko–Pastur filtering (Section 5.5) leaves exactly four eigenvalue modes above the band, carrying 25.6% of total variance. The other 74.4% is what 91 respondents answering 60 items generate by chance. Any claim resting on a single correlation, or on fine structure of the full matrix, is unsupported — which is why every downstream network is built from the denoised matrix rather than the raw one.

2. The community structure does not survive its own null. This is the sharpest negative result, and which null is used decides it. A correlation-derived network has transitivity built into its construction: if A correlates with B and B with C, A and C are pulled toward correlation automatically, and that alone manufactures modularity in structureless data. Rewiring and Erdős–Rényi nulls take the finished graph as given, so neither can detect structure the construction itself created. Only a column-permutation null — destroying association in the raw responses, then regenerating the entire chain of centre, correlate, threshold, measure — puts the claim actually being made under test. The three nulls disagree sharply:

Null model	Q_{null}	z
Degree-preserving rewiring	$0.333 \pm .013$	12.13
Erdős–Rényi	$0.420 \pm .016$	4.46
Column permutation	$0.629 \pm .044$	−3.14

Against the graph-level nulls the observed $Q = 0.491$ looks overwhelming, at z up to 12. Against the null that reproduces the construction it inverts: scrambled data pushed through the identical pipeline yields more modular graphs than the real data. Had we stopped at the conventional rewiring null we would have reported a strong, entirely spurious finding. The communities are real as a description of the graph; they are not evidence about the cohort.

3. The dimension count depends on who you keep. Filling the same 242 cells by column mean, column median or ordinal regression all yield exactly four modes (24.7%, 25.0%, 25.6% of variance) — the structure is a property of the data, not of the imputer. But dropping the ten respondents with more than twelve missing items and filling the rest with a column statistic gives only three modes at 19.3%. The count is robust to how missing cells are filled and fragile to which respondents are retained. Keeping all 91 is the better-supported choice — discarding usable respondents to avoid modelling 4.4% of cells trades away more than it protects — but the result is conditional on it.

7 Conclusion

We built an end-to-end pipeline from a raw ordinal questionnaire to a signed item network, measuring each intervention rather than asserting it. Ordinal-regression recovery of the 242 missing cells beats a modal baseline under both masking mechanisms, and row-centring eliminates 87% of the item pairs correlated at $|r| \geq 0.30$. A permutation null puts the noise floor at 0.106 against an observed spread of 0.139, Marchenko–Pastur filtering leaves four significant modes carrying 25.6% of variance, and the threshold came from a 46-point sweep by a stated rule at $|r| > 0.22$.

On the resulting network, the questionnaire’s own thematic design proves statistically real ($z = 13.4$), yet the 15 detected communities nonetheless cut across the four topic blocks (NMI 0.205).

The same framework produced the most useful negative result. That community structure clears the rewiring and Erdős–Rényi nulls but fails the column-permutation null — a finding we would have missed entirely had we used only the conventional null, and the clearest argument in this report for choosing a null that reproduces the construction. What survives is a cohort in superficial consensus with genuine but local structure: four modest cross-cutting dimensions and communities that ignore the survey’s own thematic design. Not two camps.

8 Individual Contribution

Authorship is verifiable from git log, where every commit carries its author; the per-task log with commit hashes is in CONTRIBUTIONS.md in the repository.

Member	Contribution
Arijeet Paul	Data sanitation and missingness analysis; ordinal-regression imputation and its artificial-masking validation; row-centring; src/preprocessing/ and src/viz/; Graphs 1, 2, 3 and 8.
Shreyash Lohare	Noise-floor analysis and the permutation-null framework (src/common/nulls.py) that the community analysis is tested against; Marchenko–Pastur spectral denoising and the denoised correlation matrix; the threshold sweep and its selection rule; shared configuration and analysis matrix (src/common/, src/analysis/); repository structure and run_all.py; Graphs 4, 5 and 7.
Dev Patel	Community detection on the item network and the Louvain / Leiden / Girvan–Newman comparison; null-model evaluation of the item partition (Graph 9); the reordered MP-denoised correlation heatmap (Graph 6); src/communities/; report sections for Graphs 6 and 9.